



**POLITECNICO
MILANO 1863**

Selected Topics in Music and Audio Engineering Music Instrument Classification

Francesco Crociani, Giorgio Mattina, Francesco Saverio Nisoli, Luca Trapella

June, 2026

1 Introduction

This report presents the work carried out for the Music Instrument Classification project of the Selected Topics in Music and Acoustic Engineering course. The task consists in automatically recognising which musical instruments are present in an audio signal, studied in two complementary scenarios: classification of isolated single-instrument stems (Scenario A, single-label) and classification of polyphonic mixtures where multiple instruments co-occur (Scenario B, multi-label).

The dataset used is MedleyDB [1], a collection of professionally recorded multitrack songs. The approach adopts the Audio Spectrogram Transformer (AST) [2] as the core model, and conducts a structured ablation study over three axes: fine-tuning strategy, input feature representation, and data augmentation.

2 Project Motivation

Automatic music instrument recognition is a fundamental problem in Music Information Retrieval (MIR). The ability to identify which instruments are active in an audio signal is a prerequisite for many downstream tasks, including source separation, score alignment, and music recommendation.

Despite significant advances in audio classifica-

tion driven by deep learning, instrument recognition in polyphonic mixtures remains a challenging open problem, particularly under realistic conditions where instruments overlap spectrally and temporally.

The main objective of this project is to investigate how different design choices (input feature representation, fine-tuning scope and data augmentation) affect the performance of a state-of-the-art pretrained model when transferred to a specific target dataset, namely a curated subset of MedleyDB for music instrument classification. Given the recent success of Transformer-based architectures across a wide range of audio understanding tasks, we selected a model from this family as our backbone. We aim to quantify the individual contribution of each design choice through controlled, single-variable ablation experiments, providing insights that are relevant not only to the target dataset but transferable to broader audio classification settings.

To this end, the study is structured around two complementary scenarios. Scenario A addresses single-label classification of isolated instrument stems, where exactly one instrument is active per segment and the task is well aligned with the original design of AST as a single-label classifier. Scenario B addresses multi-label recognition from polyphonic mixtures, where multiple instruments co-occur within the same audio segment. A

secondary objective of the project is therefore to assess whether a model originally conceived and pre-trained for single-label audio event classification can be effectively repurposed for the inherently harder multi-label setting, and to understand how the performance gap between the two scenarios evolves as a function of the design choices under investigation.

3 Methodology Overview

3.1 Baseline Approach

The selected state-of-the-art model is the **Audio Spectrogram Transformer** (AST) [2], the first convolution-free, purely attention-based model for audio classification. AST processes a 2D log-Mel spectrogram by splitting it into 16×16 patches with overlap, linearly projecting each patch to a 768-dimensional embedding, and feeding the sequence to a 12-layer Transformer encoder with 12 attention heads and approximately 87 million parameters. A prepended [CLS] token aggregates the sequence and is passed to a linear classification head.

The pretrained checkpoint (MIT/ast-finetuned-audioset-10-10-0.4593) was used: this model was first initialised from DeiT [3] weights (pretrained on ImageNet) and subsequently fine-tuned on the full AudioSet, achieving a single-model mAP of 0.459. AST was chosen for the following reasons:

- State-of-the-art performance on multiple audio classification benchmarks (AudioSet, ESC-50, Speech Commands).
- Accepts arbitrary 2D spectrograms as input, enabling direct replacement of the feature representation without any architectural change.
- Compact, well-documented architecture readily available on HuggingFace.
- Feasible on free-tier Colab T4 GPUs (15 GB VRAM) with a batch size of 16.

The baseline system corresponds to the **zero-shot evaluation** of the pretrained AST model (Experiment A0). In this configuration, no weights are updated during training: the AudioSet-pretrained backbone is kept entirely frozen, and only the final classification head is randomly reinitialised to map to the target instrument classes. The head is evaluated without any fine-tuning.

This setup is deliberately simple and quick to implement, and serves as a lower bound on task performance. It tests whether the general-purpose audio representations learned from AudioSet (a large-scale, 527-class audio event dataset) are directly informative for music instrument identity without any adaptation. As a reference, the zero-shot approach requires no labelled training data from the target domain and no gradient updates.

Starting from this backbone, the project investigates how three key design choices influence classification performance on the target dataset on both scenarios: the scope of fine-tuning (how much of the network to adapt, and which components to update), the input feature representation (whether Mel spectrograms, CQT, or MFCCs best suit the task), and data augmentation strategies (whether techniques such as SpecAugment and mixup provide measurable gains under a frozen backbone).

3.2 Fine-tuning AST

To adapt the pretrained backbone to the target dataset, we explored two lightweight fine-tuning strategies, chosen to span a spectrum of parameter efficiency while minimising the risk of catastrophic forgetting of the AudioSet representations.

Linear Probe. Only the classification head (`mlp_head`, $\approx 7,688$ trainable parameters) is updated, while the entire Transformer backbone remains frozen. This strategy tests whether the feature space learned during AudioSet pretraining is already sufficiently expressive for the target task, and serves as a strong parameter-efficient baseline.

BitFit + LayerNorm tuning. In addition to the classification head, the bias terms and LayerNorm parameters of all 12 Transformer blocks are unfrozen ($\approx 130K$ trainable parameters). This approach distributes a small number of updates across the full depth of the network, allowing lightweight task-specific recalibration of every layer without modifying the attention weights. It was chosen as a middle ground between a fully frozen backbone and a more expensive full fine-tuning regime, and has been shown to be surprisingly effective in low-data settings [4].

Both strategies are compared against the zero-shot baseline (A0) under a fixed Mel-spectrogram input, so that the effect of fine-tuning scope can be isolated from other design choices. The strategy

that achieves the best macro-F1 on the validation set is then adopted as the reference configuration S^* for all subsequent experiments.

For **Scenario A** (single-label), the head uses a softmax output with cross-entropy loss. For **Scenario B** (multi-label), the head is replaced by a sigmoid output with binary cross-entropy loss, allowing independent per-class predictions.

3.3 Input Feature Representations

Four input representations are compared:

Mel spectrogram. the exact configuration used during AST pretraining: 128 log-Mel filterbank bins, 25 ms Hamming window, 10 ms hop. This is the reference input for all fine-tuning and augmentation experiments.

CQT. Constant-Q Transform with 96 frequency bins over 8 octaves, bilinearly resized to 128 bins. The logarithmically spaced frequency axis mirrors the perceptual organisation of musical pitch and provides a resolution that increases towards lower frequencies, where harmonic content is denser. This makes CQT a natural candidate for instrument classification, where timbre and pitch relationships are musically meaningful. Comparing CQT against the Mel baseline allows us to assess whether a representation more closely aligned with musical structure offers any advantage over a perceptually motivated but linearly interpolated frequency axis.

MFCC-128. 128 Mel-frequency cepstral coefficients, yielding (1024, 128) arrays directly without interpolation. MFCCs are a compact cepstral representation that captures the spectral envelope of a sound, which is closely related to timbre and therefore relevant to instrument identification. Using 128 coefficients preserves fine-grained spectral detail while maintaining full compatibility with the AST input format without any geometric transformation, making it a clean ablation point.

MFCC-40. 40 MFCC coefficients (a standard configuration in speech and audio classification), with the coefficient axis linearly interpolated from 40 to 128 to match the AST input format. This variant reflects conventional practice in audio classification pipelines and allows us to evaluate whether reducing the number of coefficients to the typical operational range degrades performance once the representation is fed into a Transformer backbone. The comparison between MFCC-128 and MFCC-40 iso-

lates the effect of cepstral resolution from all other variables.

Studying these last three representations alongside the Mel baseline is particularly interesting because they encode complementary aspects of the audio signal: Mel spectrograms retain full spectral energy, CQT emphasises pitch structure, and MFCCs summarise the spectral envelope in a decorrelated, compact form. By holding the model architecture and fine-tuning strategy fixed across all variants, any difference in classification performance can be directly attributed to the input representation.

3.4 Data Augmentation

Data augmentation is a critical consideration in any supervised learning pipeline, and its importance is amplified when the available training set is small. Our subset of MedleyDB contains only 200 stems, with 25 examples per class (see Experimental Setup Section), making the model potentially susceptible to overfitting. Augmentation can act as an implicit regulariser, encouraging the model to learn representations that are invariant to superficial transformations of the input, and may therefore improve generalisation to unseen examples.

Both techniques investigated here (SpecAugment and Mixup) are already employed in the original AST training pipeline on AudioSet [5, 6]. However, there are important reasons to re-evaluate them in our setting. First, AudioSet contains over two million clips, a regime where augmentation aids generalisation but the model is unlikely to be severely data-starved. Our dataset is orders of magnitude smaller, which may change the relative benefit of each technique. Second, in our setup the backbone is kept frozen (under S^* , the linear probe strategy), meaning that augmentation only affects the training signal for the classification head rather than the full network. This is a fundamentally different scenario from the full fine-tuning regime used in the AST paper, and it is not obvious a priori whether augmentation still provides gains when the representational capacity being trained is so limited. Studying these effects in our controlled setting thus provides complementary evidence to the original results.

All augmentation experiments fix the input representation to Mel spectrograms and the fine-tuning strategy to S^* , so that any change in performance

can be attributed solely to the augmentation policy.

SpecAugment [5]. SpecAugment applies structured masking directly to the input spectrogram during training: rectangular blocks of consecutive time frames and frequency bins are zeroed out, forcing the model to rely on the remaining context for classification. We apply time masking (max 96 frames) and frequency masking (max 24 bins), set at approximately half the values used in the original AST paper to avoid over-regularisation on a dataset this small. SpecAugment is particularly relevant for instrument classification because it encourages robustness to partial occlusions of timbral and temporal cues.

Mixup [6]. Mixup generates synthetic training examples by linearly interpolating between pairs of spectrograms and their labels: $\tilde{x} = \lambda x_a + (1-\lambda)x_b$, with the composite loss $\lambda \cdot \text{CE}(y_a) + (1-\lambda) \cdot \text{CE}(y_b)$ and $\lambda \sim \text{Beta}(0.5, 0.5)$. Rather than masking existing content, Mixup implicitly augments the dataset by creating convex combinations of samples, which can smooth the decision boundary and reduce over-confidence. It also introduces a form of implicit data balancing since mixed examples draw from multiple classes simultaneously.

SpecAugment + Mixup. Both augmentations are applied jointly. Studying this combined condition is meaningful because SpecAugment and Mixup operate through complementary mechanisms (one by suppressing parts of a single input, the other by blending two inputs together) and their interaction is not straightforward to predict. A joint experiment allows us to assess whether their effects are additive, redundant, or even detrimental when the training set is small and the backbone is frozen.

4 Experimental Setup

4.1 Dataset: MedleyDB

MedleyDB [1] is a dataset of royalty-free multi-track recordings developed for annotation-intensive MIR research, containing 122 songs spanning genres including rock, pop, jazz, classical, world, and singer/songwriter. For each track it provides a stereo mix, individual processed stems, and per-stem instrument activation confidence annotations. Audio files are stereo WAV at 44.1 kHz; the dataset is released under a Creative Commons Attribution–NonCommercial–ShareAlike 3.0 license.

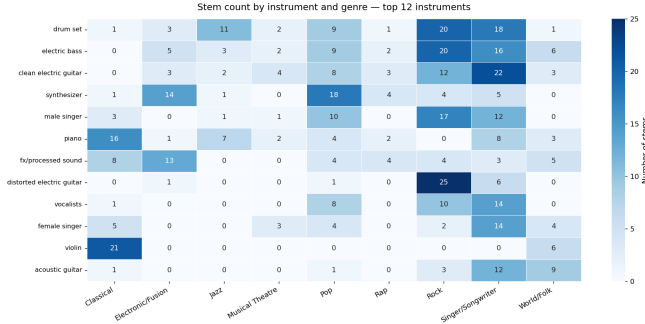
Table 1: General statistics of the full MedleyDB.

Property	Value
Total tracks	122
Total stems	856
Unique instruments	82
Genres	9
Tracks with bleed	25 (20%)
Instrumental tracks	59 (48%)
Short excerpts	17 (14%)
Stems per track (mean)	7.0
Stems per track (median)	5

Working with the full MedleyDB collection on a resource-constrained environment presents a practical challenge: the complete audio data exceeds 100 GB, making repeated loading from Google Drive prohibitively slow on a Colab T4 instance. Training directly on the full dataset would require hours per epoch, precluding the iterative experimentation that our ablation study demands. So we constructed balanced subsets for both scenarios, keeping the pipeline computationally feasible while maintaining scientific validity.

Both subsets were designed to satisfy the following shared criteria: (i) **class balance** — each target instrument class is represented by the same number of examples, so that performance differences across configurations cannot be attributed to imbalance; (ii) **genre diversity** — the selection maximises the number of genres covered per class, reducing the risk that the model learns genre-correlated cues rather than timbral ones; (iii) **track-level splitting** — the train/validation/test partition is performed at the track level, so that all stems or segments from the same original recording belong to the same split, preventing data leakage; (iv) **activation-confidence-based labels** — instrument activity is derived from MedleyDB’s per-stem activation confidence annotations rather than from coarse track-level metadata, ensuring that labels reflect actual instrument presence.

Figure 1: Stem count by instrument and genre for the 12 most frequent instruments in MedleyDB. Instruments with broad genre coverage are preferred for subset selection to reduce genre-correlated bias; *fx/processed sound* and *vocalists* were excluded due to label ambiguity despite their high stem count.

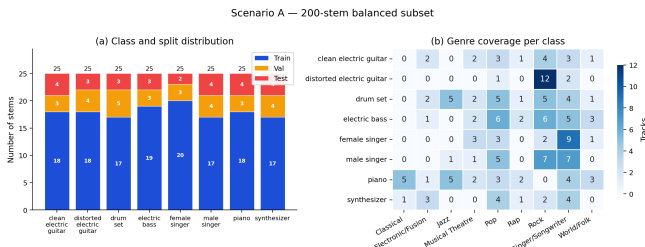


4.1.1 Subset — Scenario A (Stems)

A balanced subset was constructed from metadata without loading any audio. Only bleed-free stems were considered to ensure clean single-label annotations; instruments with fewer than 25 qualifying bleed-free stems were excluded, as were ambiguous labels such as *vocalists* and *fx/processed sound*. The eight target classes retained are: **drum set**, **electric bass**, **clean electric guitar**, **distorted electric guitar**, **piano**, **synthesizer**, **male singer**, and **female singer**.

Track selection followed a greedy strategy that prioritised tracks introducing genres not yet covered in the growing subset, enforcing exactly 25 stems per class and maximising genre diversity simultaneously. The final subset contains **200 stems** (25 per class) drawn from a pool of bleed-free tracks. The train/validation/test split was performed at track level using a grouped-stratified strategy (group = track, stratification target = instrument), resulting in **144 / 29 / 27** stems respectively, with all eight classes present in every split.

Figure 2: Scenario A — Balanced stems subset (200 stems, 8 classes). *Left*: class distribution across train, validation, and test splits; each class contains exactly 25 stems, confirming perfect balance. *Right*: number of tracks per class and genre; non-zero entries across multiple genres for every class indicate that genre diversity was successfully enforced during greedy track selection.

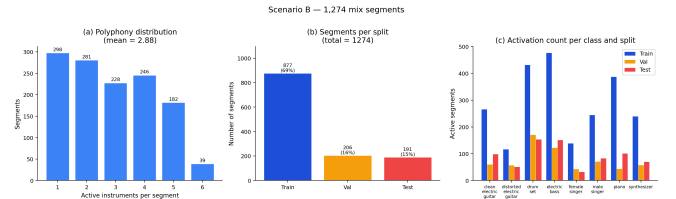


4.1.2 Subset — Scenario B (Mix)

For the polyphonic mixture task, the same set of tracks selected for Scenario A was used as source material, preserving the shared train/val/test track partition and thus preventing any cross-scenario leakage. Each full mix was divided into non-overlapping 10.24-second segments, matching the AST pretraining input length (1024 frames at a 10 ms hop). Instrument activity per segment was determined from MedleyDB activation confidence annotations: a target instrument was marked active when at least one of its stems exceeded a confidence threshold of 0.5 for at least 20% of the activation frames within that segment. No track-level fallback was used; segments whose labels could not be derived from activation confidence annotations were discarded, ensuring that all labels are annotation-grounded. Segments with no active target instrument were likewise discarded.

The resulting dataset contains **1,274 labelled segments** (877 / 206 / 191 for train / validation / test), with an average polyphony of 2.88 active instruments per segment.

Figure 3: Scenario B — Mix segment subset (1,274 segments, 10.24s each). (a) polyphony distribution showing the number of segments by count of simultaneously active target instruments; the mean polyphony of 2.88 confirms that most segments contain two to three overlapping instruments. (b) segment count per split (877/206/191 for train/validation/test), reflecting the 70/15/15 track-level partition. (c) per-class activation counts across splits, derived from MedleyDB activation confidence annotations; the distribution is consistent across splits, with no class absent from any set.



4.2 Training Procedure

To ensure that performance differences across experiments are attributable solely to the design choice under investigation, all runs share an identical training procedure. The backbone is always reinitialised

from the AudioSet pretrained checkpoint; no experiment inherits weights from a previously fine-tuned model.

Input features are normalised as $(x - \mu)/(2\sigma)$, following the convention used in the original AST paper, where μ and σ are computed exclusively on the training split of each dataset. This avoids any information leakage from the validation or test sets into the normalisation statistics.

The model is trained with AdamW and a cosine annealing learning rate schedule over a fixed number of epochs. Model selection is performed on the validation set using macro-F1 as the criterion; the best checkpoint is reloaded before final evaluation on the held-out test set. Mixed-precision training (AMP) is enabled throughout to reduce memory usage on the Colab T4 GPU. All experiments are run with a fixed random seed applied at three canonical points: before configuration, before DataLoader creation, and before model initialisation.

Table 2: Shared hyperparameters across all experiments.

Hyperparameter	Value
Epochs	20
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
LR scheduler	Cosine annealing
Training batch size	12
Evaluation batch size	16
Loss (Scenario A)	Cross-entropy
Loss (Scenario B)	Binary cross-entropy
Input shape	(1024, 128)
Normalisation formula	$(x - \mu)/(2\sigma)$
Normalisation statistics	Training split only
Model selection criterion	Validation macro-F1
Mixed precision (AMP)	Enabled
Random seed	42

4.3 Evaluation Metrics

The primary evaluation metric for both scenarios is **macro-F1**, the unweighted mean of per-class F_1 scores computed on the held-out test set. Macro-averaging treats all eight instrument classes equally and is robust to the minor class imbalances introduced by track-level splitting.

In **Scenario A** (single-label stems), audio is divided into non-overlapping 10.24-second segments during preprocessing, and evaluation is performed

at two levels. Segment-level macro-F1 is the primary metric: each segment is classified independently and performance is measured over all test segments. As a complementary metric, stem-level macro-F1 is computed by aggregating per-segment predictions for each stem via majority vote, yielding one prediction per original stem. This is a stricter measure of whether the model correctly identifies the instrument over the full active region. Supporting metrics include overall accuracy, weighted-F1, per-class precision and recall, and a row-normalised confusion matrix.

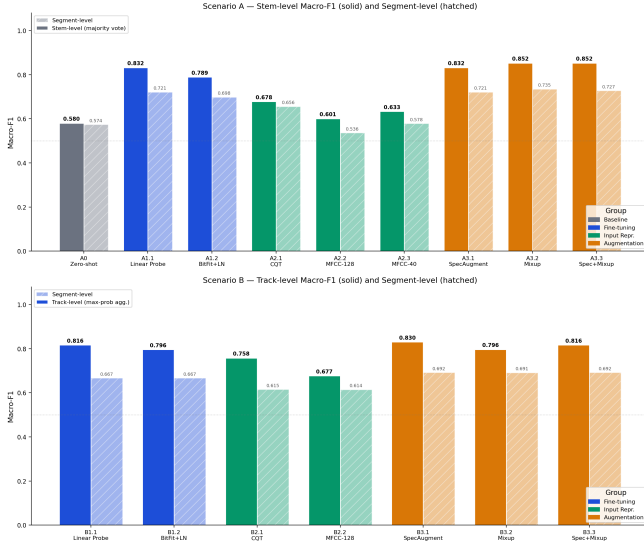
In **Scenario B** (multi-label mixtures), each 10.24-second mix segment is already the atomic unit of classification, so no aggregation step is required. The F_1 score is computed per class by treating each instrument as an independent binary prediction, and macro-F1 is the unweighted average across the eight classes. Supporting metrics include per-class precision and recall and overall subset accuracy.

In both scenarios, the validation macro-F1 is used as the checkpoint selection criterion, and final results are reported by reloading the best validation checkpoint and evaluating on the test set.

5 Evaluation and Results

All experiments are evaluated on the held-out test set. For Scenario A the primary metric is **stem-level macro-F1** (majority vote over segments of each stem), with segment-level macro-F1 reported as a secondary metric. For Scenario B the primary metric is **track-level macro-F1** (max-probability aggregation over segments of each track), with segment-level macro-F1 as secondary. Cross-scenario comparisons use segment-level macro-F1, the only metric common to both scenarios.

Figure 4: Primary and secondary Macro-F1 on the test set for all experiments. Solid bars show stem-level F1 (Scenario A) and track-level F1 (Scenario B); hatched bars show segment-level F1 in both cases. Colours identify the experiment group: grey = baseline, blue = fine-tuning, green = input representation, orange = augmentation.



5.1 Fine-tuning Strategy

The zero-shot baseline (A0) achieves a stem-level macro-F1 of 0.58, confirming that AudioSet representations are informative for instrument identity even without any task-specific adaptation: seven of the eight classes are recognised above chance by the frozen backbone alone. However, supervised adaptation with as few as 7,688 trainable parameters (Linear Probe, A1.1) raises stem-level macro-F1 to 0.83, a gain of +0.25 over zero-shot. This result demonstrates that the AudioSet feature space already encodes the timbral information needed for instrument classification, and that a single linear layer is sufficient to decode it.

BitFit + LayerNorm (A1.2), which updates bias terms and LayerNorm parameters across all 12 Transformer blocks ($\approx 49K$ parameters), reaches a stem-level macro-F1 of 0.79, below Linear Probe despite having six times more trainable parameters. The same ranking holds in Scenario B: Linear Probe achieves track-level macro-F1 of 0.82 versus 0.80 for BitFit + LayerNorm, a consistent gap of +0.02 across both tasks. We attribute this reversal to the small training set size (144 stems, 877 mix segments): distributing gradient updates across the full depth of the network on so few examples introduces more variance than it reduces bias, whereas a single linear head over frozen features benefits from a stable, well-regularised optimisation landscape. Linear Probe is therefore adopted as S^* for all subsequent experiments.

5.2 Input Feature Representation

With the fine-tuning strategy fixed to S^* , replacing the Mel spectrogram with alternative representations leads to consistent performance drops in both scenarios (Figure 4).

In Scenario A, the CQT (A2.1) reaches a stem-level macro-F1 of 0.68, a drop of -0.15 relative to the Mel baseline (A1.1, 0.83). MFCC-128 (A2.2) falls further to 0.60, and MFCC-40 (A2.3) to 0.63. The pattern is reproduced in Scenario B: CQT (B2.1) achieves a segment-level macro-F1 of 0.62 compared to 0.67 for the Mel baseline (B1.1), and MFCC-128 (B2.2) reaches 0.61.

The dominant factor explaining the Mel advantage is *domain matching*: AST was pretrained on AudioSet using 128-bin log-Mel spectrograms with a 25 ms Hamming window and 10 ms hop, which is precisely the representation used in our Mel experiments. When the input distribution shifts to CQT or MFCC, the backbone’s patch embedding layer, trained to interpret log-Mel patches, must generalise outside its pretraining domain. CQT performs better than MFCC, possibly because its logarithmic frequency axis is visually more similar to a log-Mel spectrogram than a cepstral representation; but neither alternative recovers the performance of the matched representation.

The notable exception is MFCC-128 in the cross-scenario comparison (A2.2 vs B2.2): it is the only configuration where Scenario B outperforms Scenario A at segment level (gap = -0.08), and both scenarios perform near the zero-shot level. We interpret this as a floor effect: when the representation is far from the pretraining distribution, neither the isolated-stem nor the polyphonic-mix setting provides enough signal to compensate, and the relatively richer training set of Scenario B provides a marginal advantage.

5.3 Data Augmentation

Augmentation experiments fix the input to Mel spectrograms and the strategy to S^* , isolating the effect of the augmentation policy.

In Scenario A, SpecAugment alone (A3.1, stem-F1 = 0.83) matches the unaugmented baseline (A1.1, 0.83) exactly at stem level, while Mixup (A3.2, 0.85) and the joint policy (A3.3, 0.85) each deliver a gain of +0.02. The result suggests that masking parts of the spectrogram does not provide

additional regularisation benefit when the backbone is fully frozen and only the linear head is trained: SpecAugment disturbs the input patches that the frozen encoder would otherwise summarise cleanly, offering little benefit to a head with so few parameters. Mixup, by contrast, creates convex combinations of training examples, effectively expanding the boundary of the training distribution visible to the head and smoothing the decision boundary. Combining both (A3.3) does not improve over Mixup alone at stem level, suggesting that the two mechanisms are redundant rather than complementary in this regime.

In Scenario B all three augmentation variants improve over the unaugmented baseline (B1.1 segment-F1 = 0.67): SpecAugment (B3.1) reaches 0.69, Mixup (B3.2) reaches 0.69, and the joint policy (B3.3) reaches 0.69. At track level, SpecAugment (B3.1, 0.83) is the strongest single variant, marginally ahead of the joint policy (B3.3, 0.82) and Mixup (B3.2, 0.80). Unlike in Scenario A, SpecAugment provides a measurable benefit in B, likely because masking temporal and frequency bands discourages the model from relying on specific spectral patterns that co-occur reliably in isolated stems but are confounded by instrument overlap in mixtures.

5.4 Per-class Analysis

Figure 5 and 6 show the per-class segment-level F1 heatmaps.

Easy classes. Drum set and electric bass are the best-recognised instruments across every configuration and both scenarios, consistently exceeding $F1 = 0.90$. Their distinctive spectro-temporal signatures (transient-dominated for drums, low-frequency harmonic for bass) are well-represented in AudioSet and transfer cleanly to our task.

Persistently difficult classes. Clean electric guitar and female singer are the hardest classes in both scenarios. Clean electric guitar is severely confused with distorted electric guitar in Scenario A (best $F1 = 0.59$ at A0, dropping to 0.04 under MFCC-128), reflecting the spectral overlap between the two timbres and the sensitivity of the distinction to the input representation. Female singer is the weakest class in Scenario B across all configurations (best $F1 = 0.34$ at B3.1), more than 0.15 points below the next-weakest class. The confusion arises from male-female vocal overlap: in polyphonic mix-

tures a female voice is frequently co-active with a male singer whose energy dominates the mixture, causing the model to systematically under-predict female singer.

Scenario A vs B error patterns. The confusion between clean and distorted electric guitar is pronounced in Scenario A, where isolated stems should in principle make the distinction easier. In Scenario B the two guitar classes actually have higher F1, suggesting that when guitars are embedded in a full mix, other co-active instruments (drums, bass) provide contextual cues that indirectly help the model regularise its predictions. Conversely, piano is easier in Scenario A ($F1$ up to 0.86 under A3.3) than in B (maximum 0.72), where its harmonic content overlaps with synthesizer and clean guitar in dense mixes.

Figure 5: Scenario A — per-class segment-level F1 heatmap (test set). Rows are experiments; columns are instrument classes. Values below 0.5 indicate classes where the model performs near or below chance.



Figure 6: Scenario B — per-class segment-level F1 heatmap (test set). Female singer is the weakest class across all configurations, consistently below $F1 = 0.35$.



Figure 7: Scenario A — Confusion matrices (test set, segment-level): drum set and electric bass are consistently well-recognised (≥ 0.94), while clean vs. distorted electric guitar and female vs. male singer are the dominant confusion pairs across all configurations.



Figure 8: Cross-scenario comparison. *Left*: segment-level Macro-F1 for A and B (common metric). *Centre*: task-native primary metrics (A stem vs B track). *Right*: gap A–B at segment level; positive values indicate configurations where the stem task is easier.

A striking feature of the Scenario B results is that segment-level macro-F1 is nearly identical across the two fine-tuning strategies (B1.1 and B1.2 both report 0.6665 at four decimal places), whereas track-level macro-F1 diverges by 0.02 in favour of Linear Probe. This phenomenon is explained by the structure of multi-label evaluation at segment level. With a fixed decision threshold of 0.5, two models with similar calibration produce nearly identical binary predictions per segment even when their output probabilities differ. Track-level aggregation via max-probability, by contrast, amplifies even small confidence differences across segments: the model that assigns consistently higher probability to the correct classes will see those classes survive the threshold more reliably after aggregation, while the weaker model’s lower-confidence predictions are more susceptible to noise. Track-level evaluation therefore reveals calibration differences that segment-level evaluation masks, and is the more discriminative metric for comparing Scenario B configurations. A similar argument applies to the augmentation experiments in Scenario B, where segment-level F1 varies by at most 0.003 across the three variants while track-level F1 varies by 0.02. This reinforces the recommendation to treat track-level macro-F1 as the primary metric for Scenario B.

Across the representation experiments, the A–B gap is largest for Mixup (+0.044) and smallest or reversed for MFCC-128 (−0.079). The inversion for MFCC-128 is consistent with the floor-effect interpretation discussed in Section 5.3: when the representation does not transfer well, the richer training set of Scenario B provides a slight compensating advantage. At the primary-metric level, the gap between the best Scenario A configuration (A3.2/A3.3 stem-F1 = 0.85) and the best Scenario B configu-

5.5 Scenario A vs Scenario B

Figure 8 compares both scenarios using segment-level macro-F1 as the common metric. For most configurations the gap is positive (Scenario A outperforms B), reflecting the additional difficulty of multi-label polyphonic recognition: instruments overlap spectrally and temporally, activation labels are probabilistic rather than binary, and the model must produce simultaneous predictions for multiple classes without any single-instrument reference signal.

ration (B3.1 track-F1 = 0.83) is modest, suggesting that a purely attention-based model pretrained on large-scale audio data can be adapted to polyphonic multi-label recognition with comparable effectiveness to single-label stem classification, provided an appropriate evaluation metric and aggregation strategy are used.

6 Conclusions

This study investigated how fine-tuning strategy, input representation, and data augmentation affect the performance of a pretrained AST model on music instrument classification, across both an isolated-stem (Scenario A) and a polyphonic-mixture (Scenario B) setting.

The most significant finding is that even minimal supervised adaptation is highly effective: a Linear Probe with only 7,688 trainable parameters

raises stem-level macro-F1 from 0.58 (zero-shot) to 0.83, confirming that AudioSet representations already encode the timbral structure needed for instrument recognition. Domain-matched input is the second critical factor: replacing the Mel spectrogram with CQT or MFCC consistently degrades performance, as the AST patch embedding is optimised for log-Mel patches. Among augmentation strategies, Mixup yields the best results in Scenario A (stem-F1 = 0.85), while SpecAugment is the strongest single variant in Scenario B (track-F1 = 0.83), where temporal and frequency masking provides additional robustness to instrument overlap. The modest gap between the best Scenario A and Scenario B configurations (0.85 vs 0.83 at the primary metric) suggests that a single-label pretrained Transformer can be repurposed for multi-label polyphonic recognition with comparable effectiveness, provided an appropriate aggregation strategy is used at evaluation time.

References

- [1] R. Bittner, J. Salamon, M. Tierney, M. Mauch, C. Cannam, and J. Bello, “MedleyDB: A multitrack dataset for annotation-intensive MIR research,” in *Proc. ISMIR*, 2014, pp. 155–160.
- [2] Y. Gong, Y.-A. Chung, and J. Glass, “AST: Audio Spectrogram Transformer,” in *Proc. Interspeech*, 2021.
- [3] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, “Training data-efficient image transformers & distillation through attention,” *arXiv:2012.12877*, 2020.
- [4] E. Ben Zaken, S. Ravfogel, and Y. Goldberg, “BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models,” in *Proc. 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, Dublin, Ireland, 2022, pp. 1–9.
- [5] D. S. Park, W. Chan, Y. Zhang, C.-C. Chiu, B. Zoph, E. D. Cubuk, and Q. V. Le, “SpecAugment: A simple data augmentation method for automatic speech recognition,” in *Proc. Interspeech*, 2019.
- [6] Y. Tokozume, Y. Ushiku, and T. Harada, “Learning from between-class examples for deep sound recognition,” in *Proc. ICLR*, 2018.